

Inverse Design and Sequential Dual-SLM Realisation of a Source-S

Source-scale technical evidence draft. Microfabrication/sample-plane success is not claimed.

This PDF is a fallback rendering because the local environment lacks a LaTeX engine. The editable LaTeX source is still written to report/nathan_hexagonal_bessel_full_report.tex.

Abstract

This report derives the source-scale branch from the Gaussian vector target through ideal Jones synthesis, inverse reconstruction, realistic carrier/4F filtering, tolerance audit, strict gate repair and the corrected sequential single-beam dual-SLM architecture.

Introduction

The project asks whether Nathan's segmented radial/azimuthal source-scale vector Bessel field can be reproduced with a programmable laboratory route. The answer at this stage is yes in simulation for the source-scale branch. The aim is not to claim microfabrication or sample-plane success. It is to establish the mathematical field, the optical transformations, the realistic carrier/4F implementation, and the corrected sequential dual-SLM bench architecture.

Physical and Mathematical Background

The source field begins as a Gaussian amplitude $A(r)=\exp(-r^2/w_0^2)$. The six-sector structure is not a scalar intensity mask. It lives in the local polarization angle $\alpha(\theta)=\theta+\phi_0(\theta)$, where ϕ_0 alternates between radial and azimuthal sectors. Thus the pre-axicon target is $E=A[\cos \alpha, \sin \alpha]^T$, and S_0 remains Gaussian before the axicon. Stokes fields reveal the polarization structure; intensity alone does not.

V0 Source-Model Reproduction

V0 established the reference observable and grid convention. The Nathan-style axis-sampled grid is required because a zero-straddling grid creates a central artefact at the angular singularity. The literal source port and the project implementation reproduce the same source-scale field. This is the reference used by later route comparisons.

Vector Axicon Propagation

After the target vector field reaches the axicon, the local radial and azimuthal components see the source-scale conical phase and p/s response. The propagated observable is total intensity $|E_x|^2 + |E_y|^2 + |E_z|^2$. The vector angular-spectrum propagator enforces transversality and advances each spectral component by $\exp(i k_z z)$. The hexagonal structure is therefore a downstream propagation result, not merely a pre-axicon drawing.

Ideal Jones-Chain Synthesis

M2P showed two analytic routes. A patterned HWP with $\beta = \alpha/2$ maps horizontal input to the target Jones vector. The dual-SLM/QWP route uses phase masks $\phi_H = +\alpha$ and $\phi_V = -\alpha + \pi/2$, followed by a QWP in the selected project convention. The derivation is power preserving; relative piston rotates the local polarization state and remains visible to polarimetry even when total intensity is insensitive.

Forward Source-Scale Bench Replication

M2N propagated the ideal patterned-HWP, ideal dual-SLM/QWP, and realistic dual-SLM + carrier + 4F + QWP routes through the same source axicon. The ideal routes reproduce V0 at $z=60$ mm. The realistic carrier/4F route keeps the strict visual hexagon with $z=60$ correlation around 0.9936 and first-order efficiency around 0.949.

Backward / Adjoint Mask Synthesis

M2Q starts from the full complex V0 target at $z=60$ mm, backpropagates it, applies an inverse axicon operation, and removes the QWP. The recovered H/V phase solution independently returns $\phi_H = +\alpha$ and $\phi_V = -\alpha + \pi/2$. This is important because it confirms the forward analytic mask convention from the inverse problem.

Realistic 4F Filtering

A displayed phase mask includes the desired phase plus a carrier $2\pi\nu x$. In the Fourier plane the +1 order is displaced by $x = \lambda f \nu$. With $\lambda = 1029$ nm, $f = 300$ mm and $\nu = 6.25$ lp/mm, the displacement is about 1.929 mm and the recommended iris diameter is about 1.54 mm. The common 4F route filters both polarization channels together.

Lab Realism and Tolerance

M2S introduced quantisation, fill factor, H/V imbalance, piston, QWP errors, iris errors, registration, axicon decentre and z offsets. Within tested ranges, most representative imperfections were relatively forgiving. The dominant practical sensitivity is hologram/mask centre relative to the axicon axis. A 0.5 mm axicon/mask offset is not acceptable blindly but can be substantially recovered by digital recentring.

High-Resolution and Sampling Audit

M2U separated numerical resolution from display resolution. $N=512$ is Nyquist-safe but not publication-preferred. $N=1024$ is useful confirmation. $N=1536$ is publication-selected for the hero comparison with about 9.89 native samples per radial fringe. M2W-FIX adds SAS-scaled focus crops so close-up panels are rendered on finer output grids while classifier metrics remain tied to native validated arrays.

Optimisation Integrity and Strict Hexagon Gate

M2U2-FIX repaired an important failure: high full-field correlation can be dominated by dark background and can hide fourfold or triangular structures. The old best compromise is permanently forbidden. The strict gate uses class, focus support, angular correlation, c60/c120 behaviour, dark-core limits and fourfold vetoes. Its 0.997 reference-correlation floor is a project-calibrated eligibility threshold, not a universal definition of a hexagon.

Native-Panel and Hardware Provenance

The panel raster is 1920 x 1080 at 8 um pitch, giving a 15.36 x 8.64 mm active area. The exact PLUTO-2.1 NIR-149 identity is externally supplied until physical labels/manuals are read. Phase stroke and LUT are calibration-required. Camera scale, z-stage and QWP mount sign also remain bench calibrations.

Sequential Single-Beam Dual-SLM Architecture

M2W-FIX supersedes the original split-arm diagram. The accepted implementation is collinear and sequential: equal H/V preparation, SLM1, optional swap HWP, SLM2, optional swap-back, common 4F, QWP, axicon and camera. The sequential Jones derivation gives $E_4 = A/\sqrt{2} [\exp(i \phi_H), \exp(i \phi_V)]^T$. The pre-QWP and post-QWP overlaps are 1.0, the ideal $z=60$ correlation is 1.0, and the realistic sequential route remains strict-eligible.

Sequential Power / Energy Ledger

The corrected power model has no split-arm H/V or PBS recombination rows. It tracks one beam through preparation, SLM1, optional swap, SLM2, common 4F selection, QWP, axicon, $z=60$ integration and useful-region power. The 1 W and 10 W examples are linear model examples only and are not damage-threshold claims.

Closed-Loop Correction

The camera observes final intensity, centre, symmetry, dark core, lobe balance and z structure. Shack-Hartmann measurements address low-order common wavefront errors. Polarimetry checks pre-axicon vector field, H/V balance, relative phase and QWP sign. The bounded correction loop improves several degraded cases, but not every failure is fully recovered to strict eligibility.

Final Lab Build

The recommended bench is a 1029 nm Gaussian source, POL/HWP preparation, sequential SLM1/SLM2 masks, common 300 mm 4F with 6.25 lp/mm carrier and 1.54 mm iris, QWP at nominal code -45 deg, 2 deg n=1.458 axicon, and a camera z scan around z=60 mm. The six-piece segmented optic is not required for this programmable source-scale trial.

Conclusions

Nathan's source-scale field has been reproduced in simulation; the propagated result is a genuine strict-gate hexagonal field; the vector field can be synthesised analytically; inverse propagation recovers the same masks; realistic common-4F filtering preserves the beam; the main practical sensitivity is mask/axicon centring; the sequential single-beam dual-SLM architecture is valid in simulation. Source-scale lab trial is justified. Microfabrication/sample-plane success is not established.

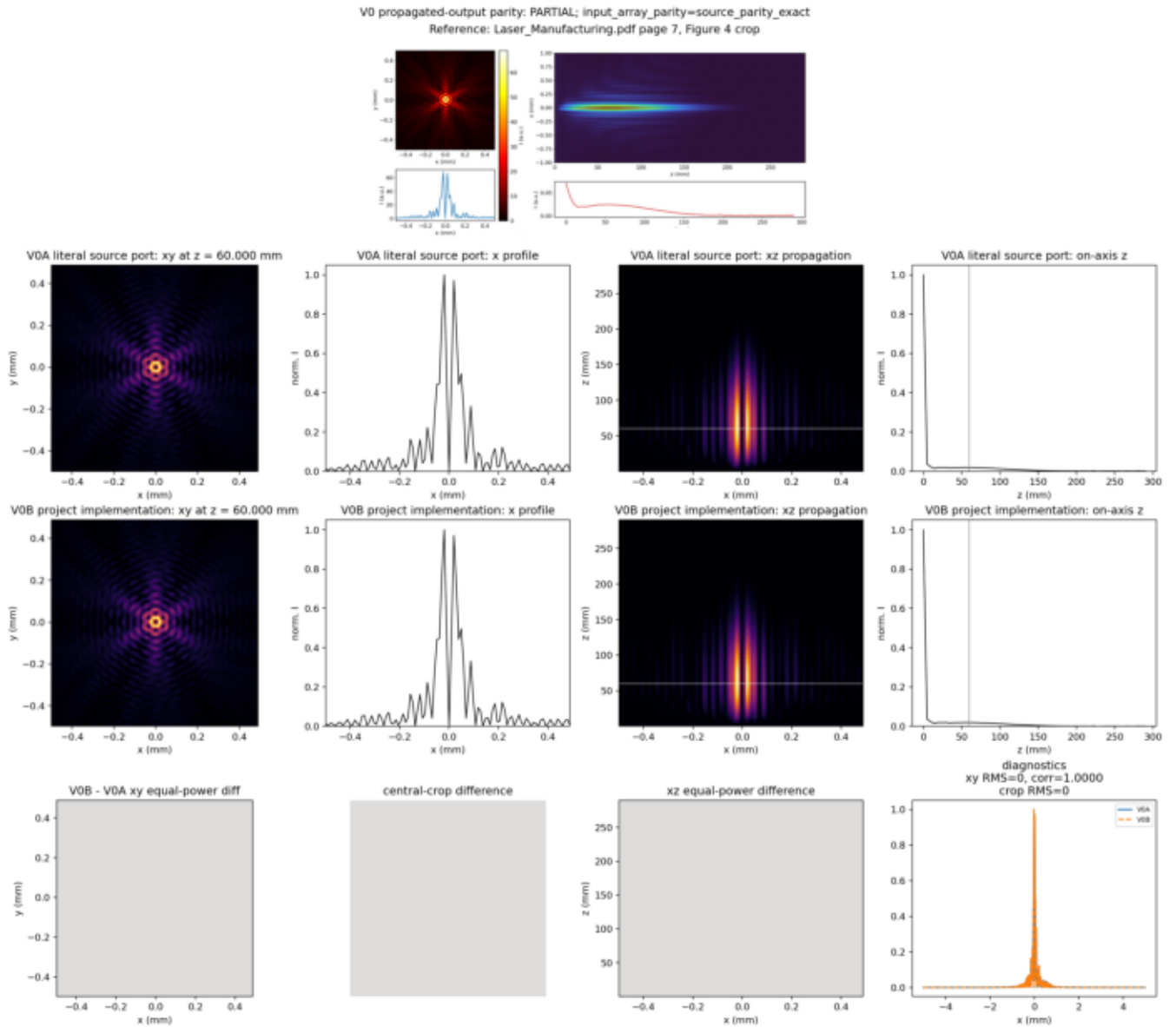
Limitations

The exact SLM phase stroke and per-panel LUT are not measured here. Camera scale and z-stage calibration remain unresolved. Exact LC-director orientation is unverified. The 4F geometry is nominal until bench measurement. Simulations do not prove experimental success. Strict thresholds are project-calibrated. No microfabrication/sample-plane success claim is made.

Next Steps

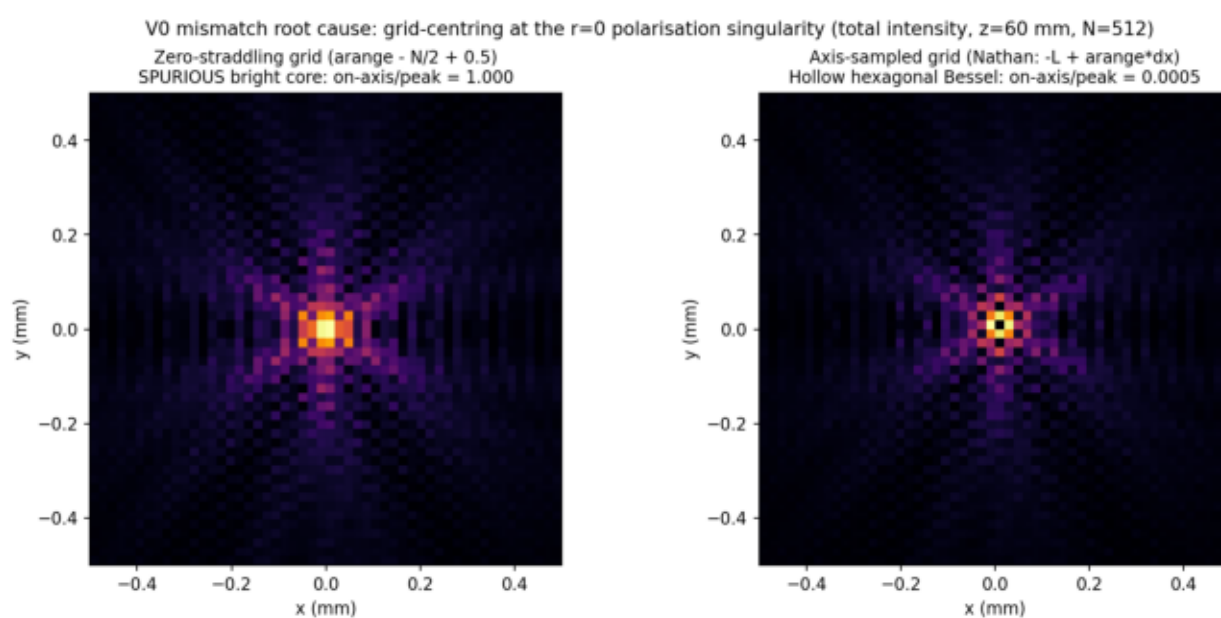
Read physical SLM labels and determine LC directors; measure phase LUTs; deploy native masks; calibrate 4F displacement and iris; validate pre-axicon Stokes fields; insert the axicon; perform camera z scans; digitally recentre; add Shack-Hartmann correction; then use a bounded camera-driven correction loop before revisiting microfabrication adaptation.

F01: V0 source parity and reference/reproduction comparison.

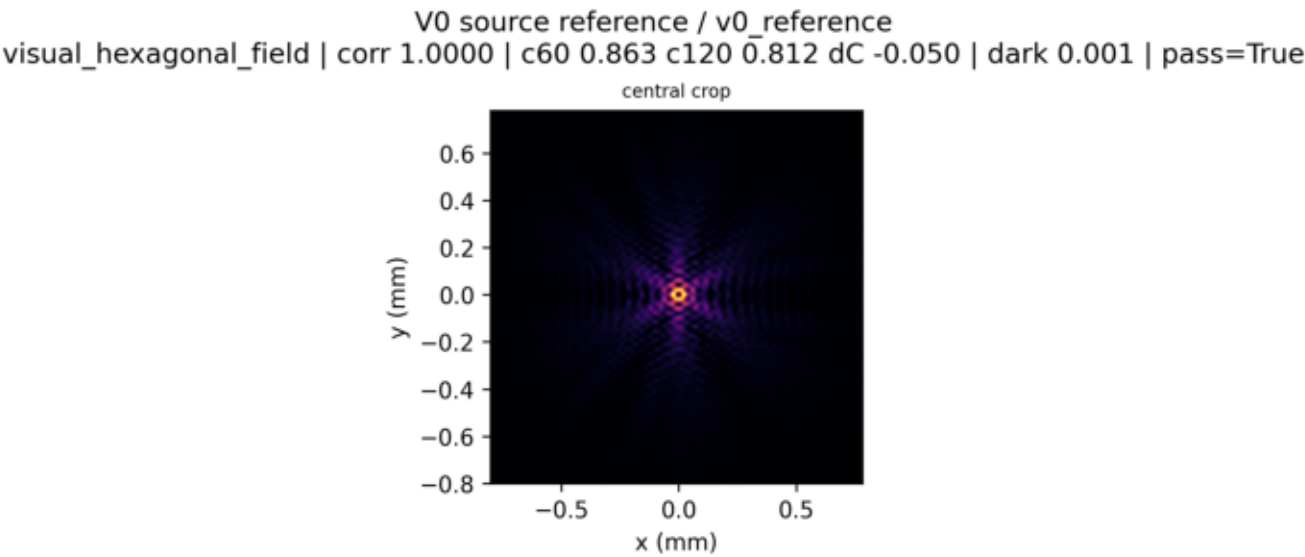


Source: outputs/figures/digital_twin/nathan_visual_ladder/nathan_visual_ladder_v0_reference_vs_reproduction.png

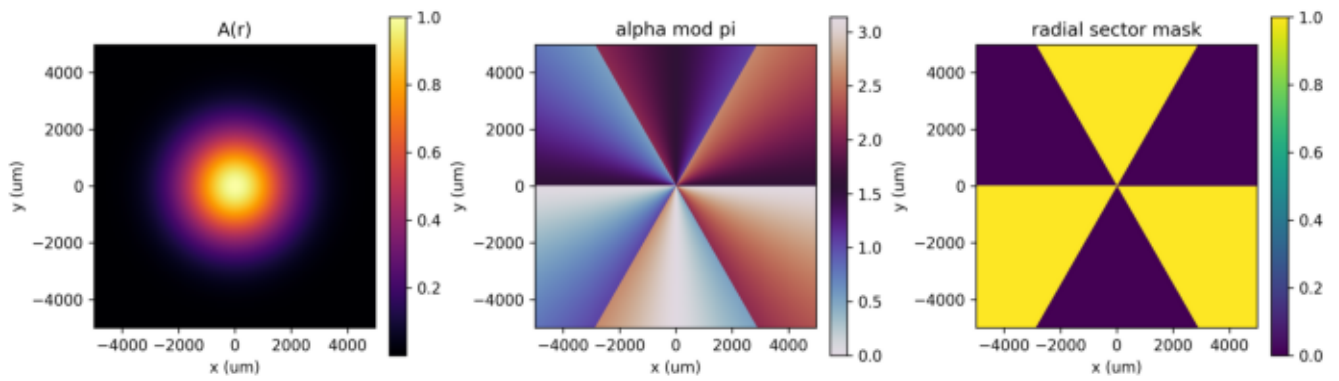
F02: Grid convention audit showing why Nathan axis-sampled centring is required.



F03: High-resolution V0 reference crop and propagated structure.

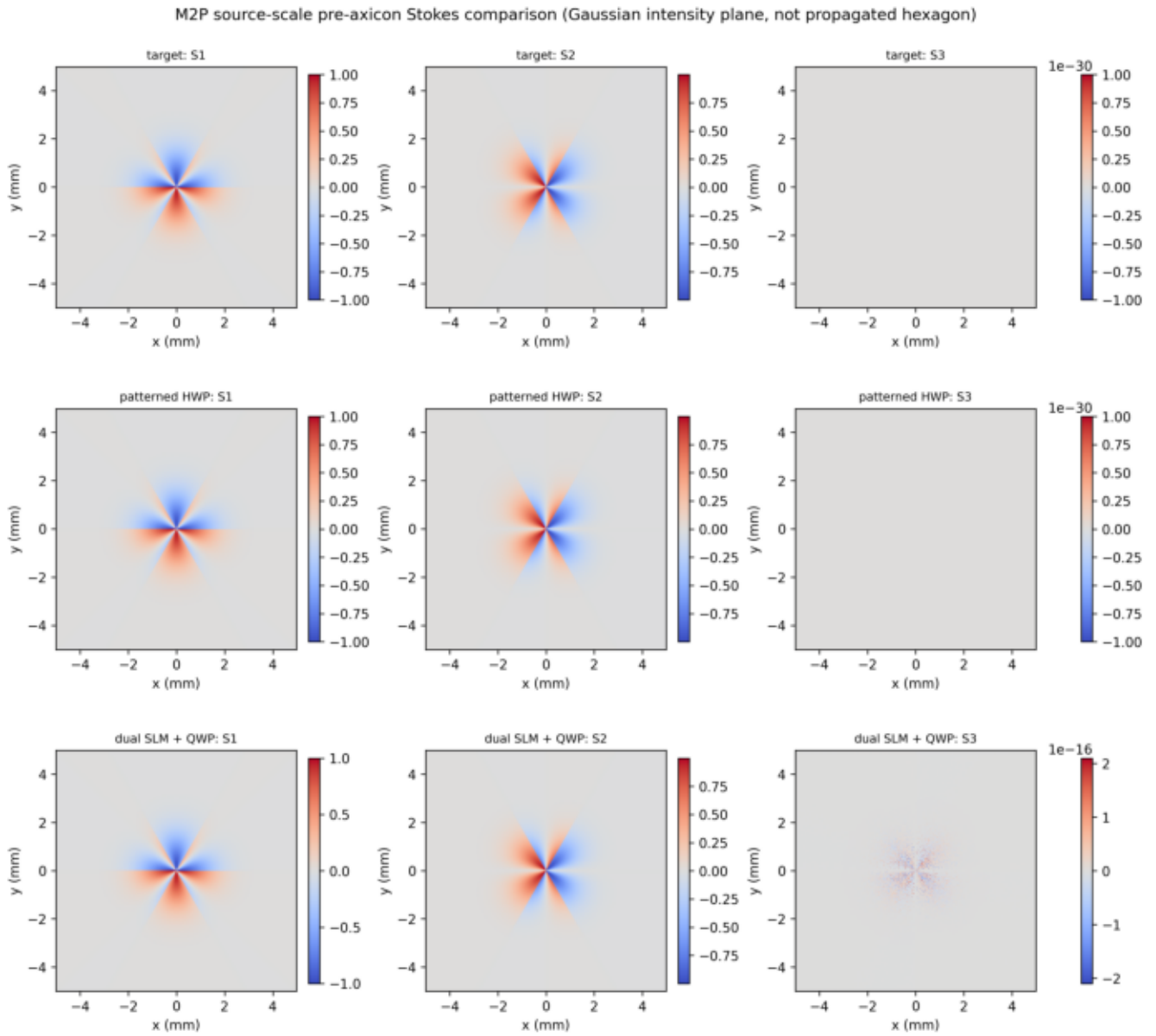


F04: M2P target alpha and sector map.

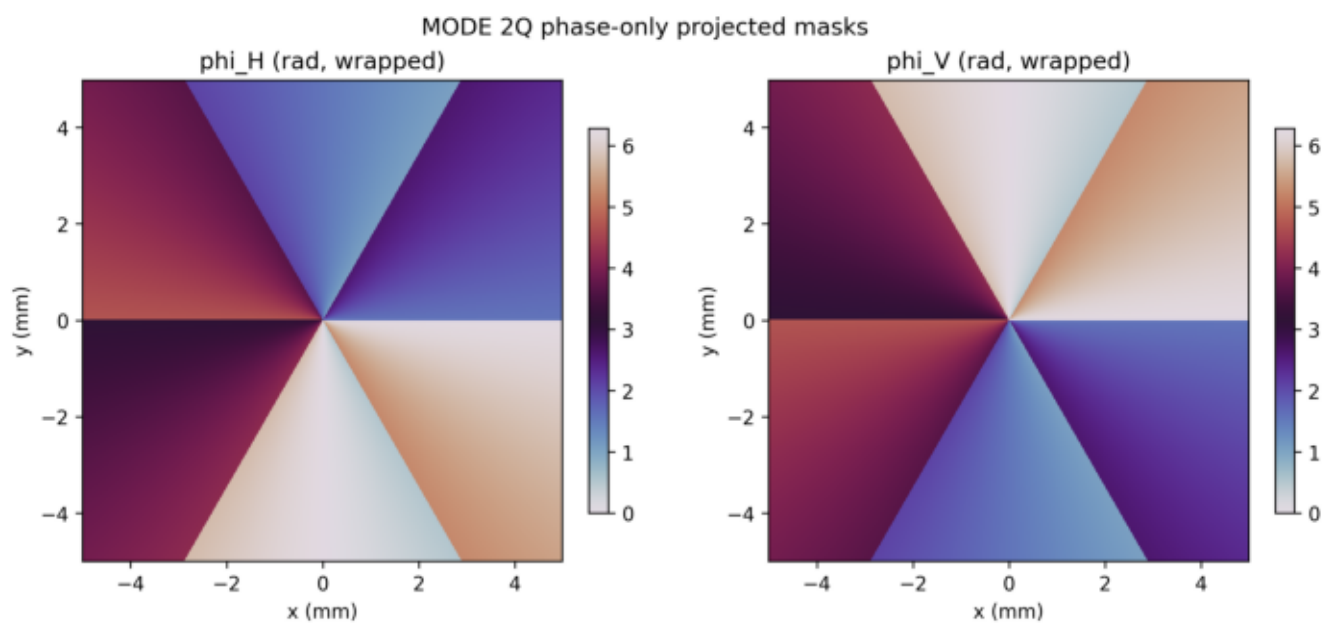


Source: outputs/figures/digital_twin/nathan_mode2u_master_highres_audit/01_m2p_preaxicon/m2p_target_alpha_sector_highres.png

F05: M2P pre-axicon Stokes comparison.

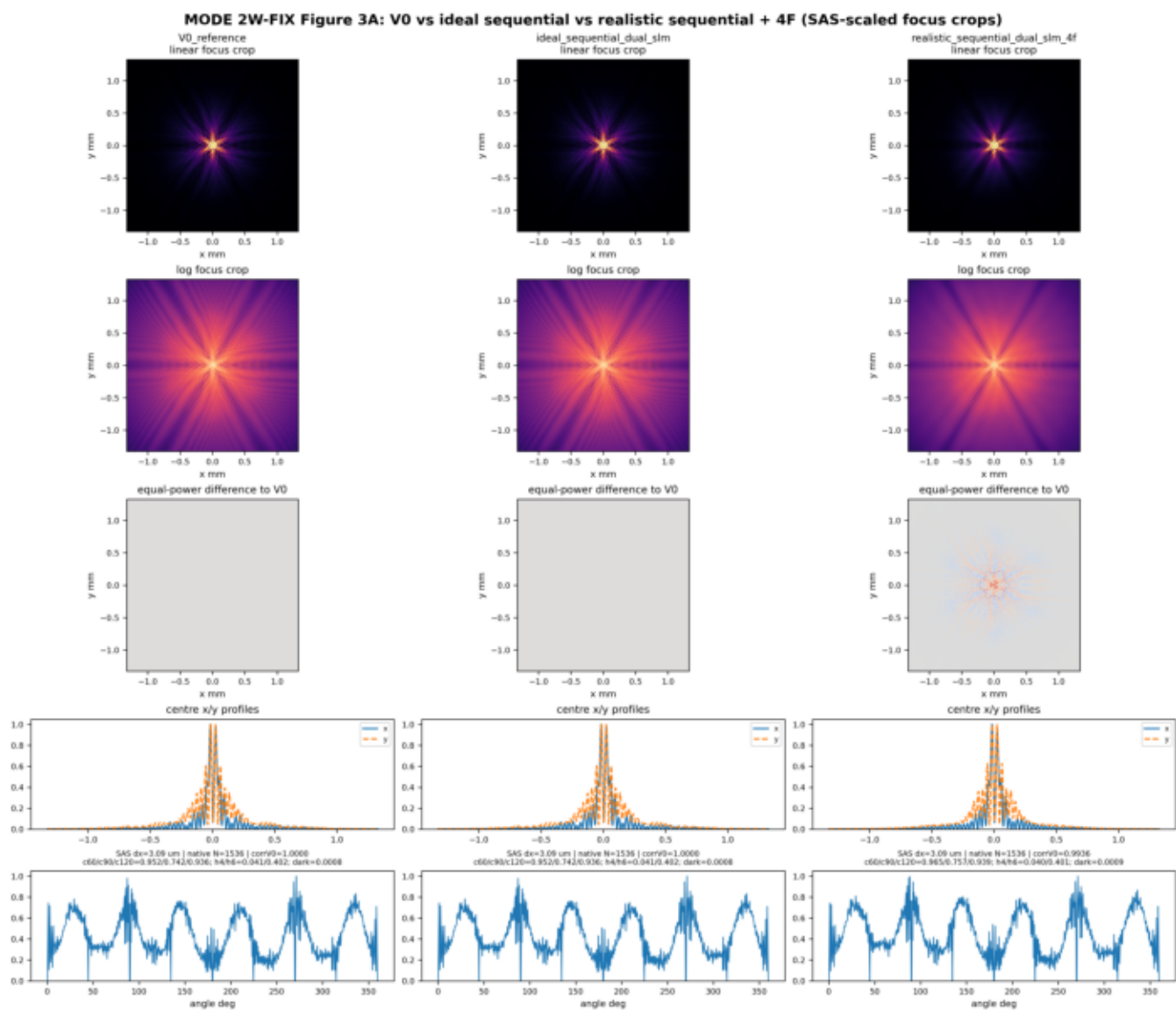


F06: M2Q recovered phase-only masks and inverse design diagnostics.



Source:
outputs/figures/digital_twin/nathan_mode2u_master_highres_audit/03_m2q_inverse_masks/m2q_phase_only_masks_highres.png

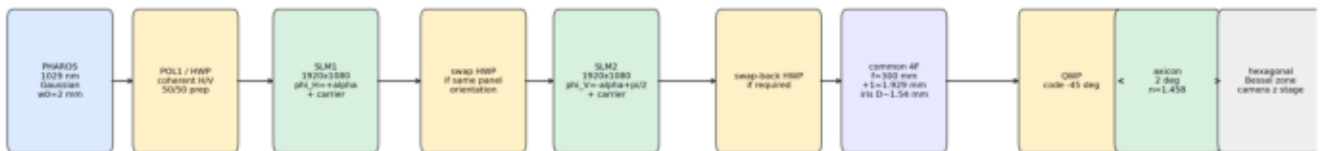
F07: M2W-FIX V0 versus ideal sequential versus realistic sequential comparison.



Source: outputs/figures/digital_twin/nathan_mode2w_fix_sequential_master/03_ideal_vs_realistic/fig3A_v0_ideal_realistic_sequential.png

F08: Corrected sequential single-beam optical system.

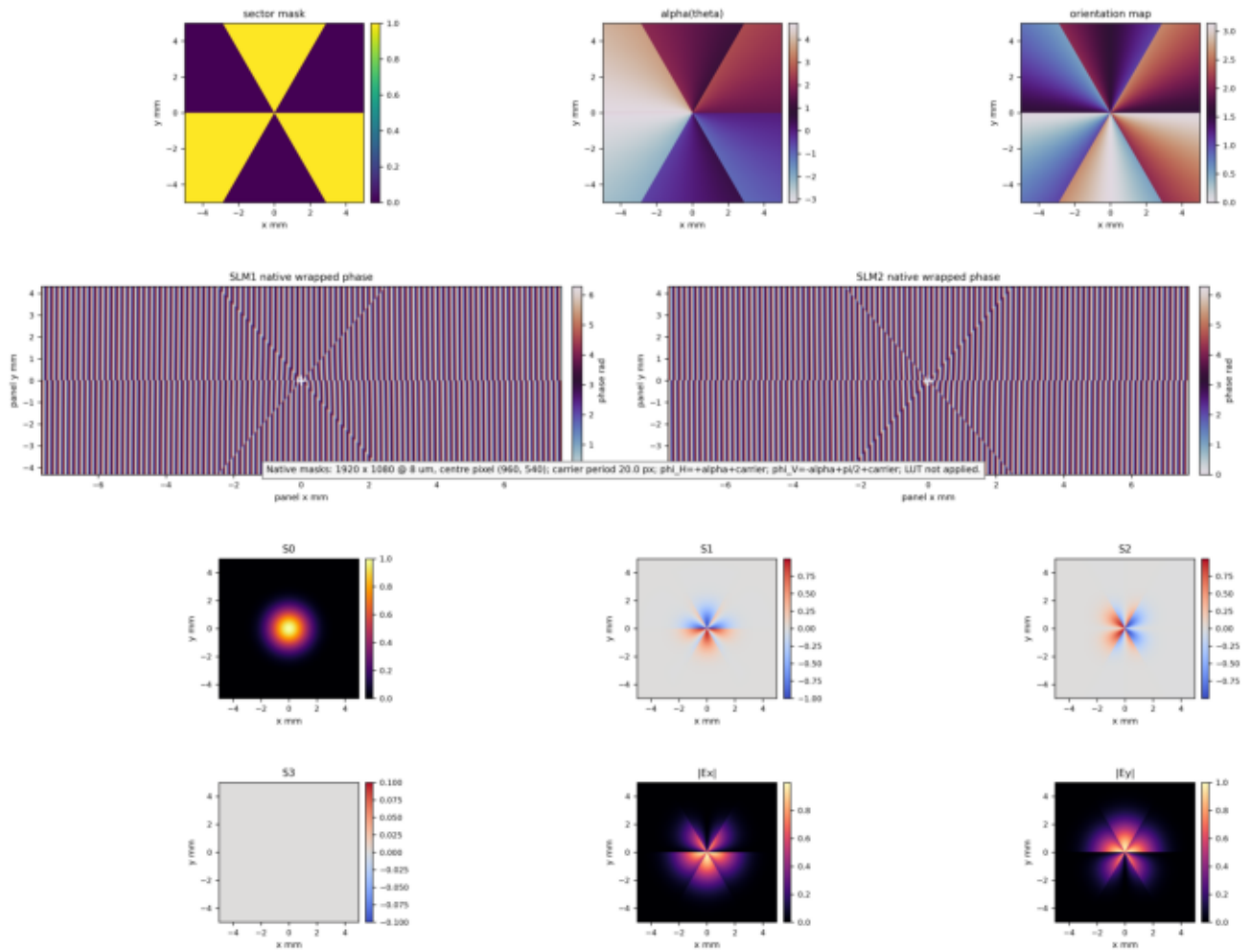
MODE 2W-FIX Figure 1: sequential single-beam architecture (no PBS split, no H/V interferometer arms)



Alternate valid branch: if SLM2 is mounted with orthogonal LC director, swap/swap-back HWPs may be omitted after the panel-orientation test.

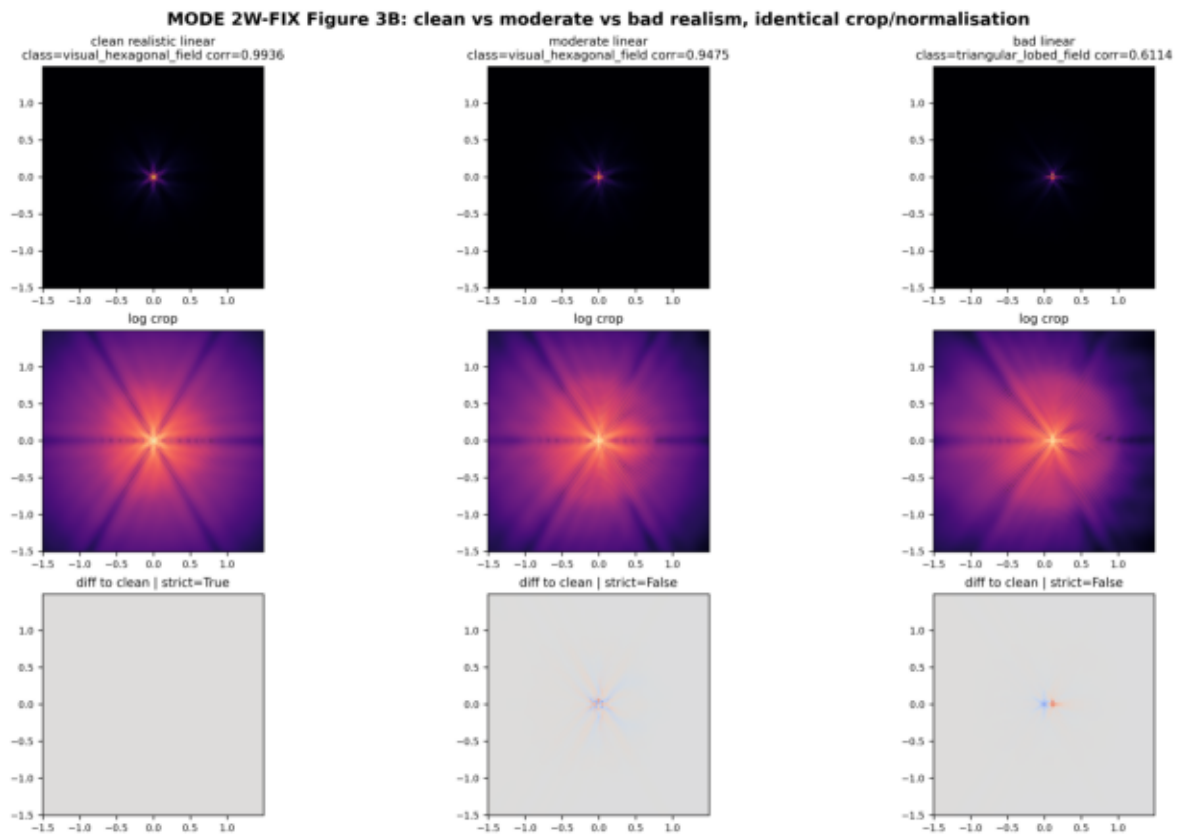
F09: Native SLM1/SLM2 masks and target vector-field diagnostics.

MODE 2W-FIX Figure 2: target vector field and native sequential SLM1/SLM2 masks; LUT not applied

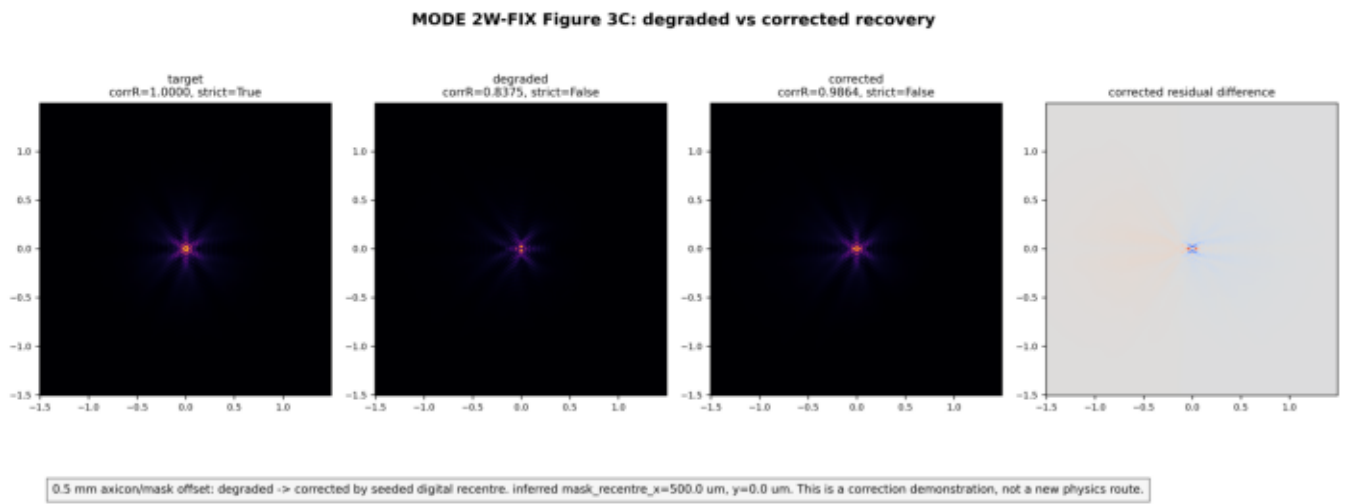


Source: outputs/figures/digital_twin/nathan_mode2w_fix_sequential_master/02_target_masks/fig2_target_and_sequential_masks.png

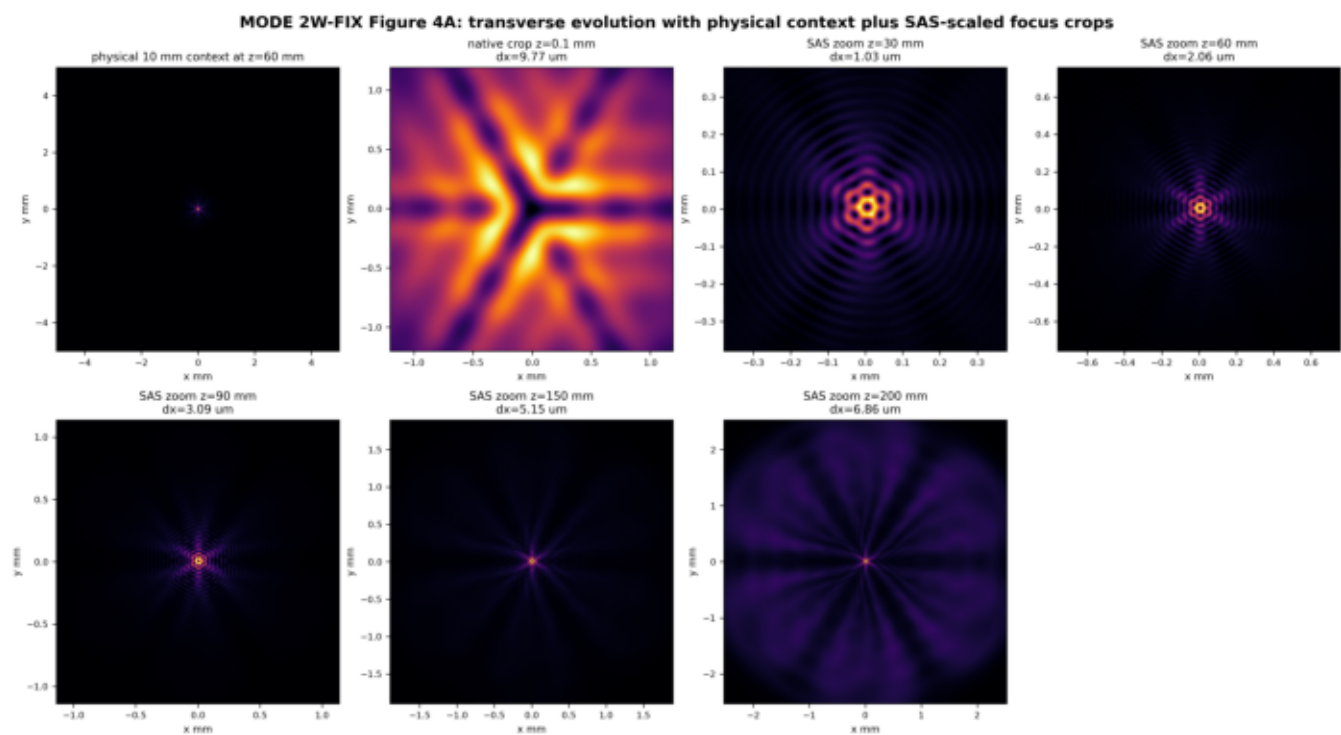
F10: Realism degradation: clean, moderate and bad cases.



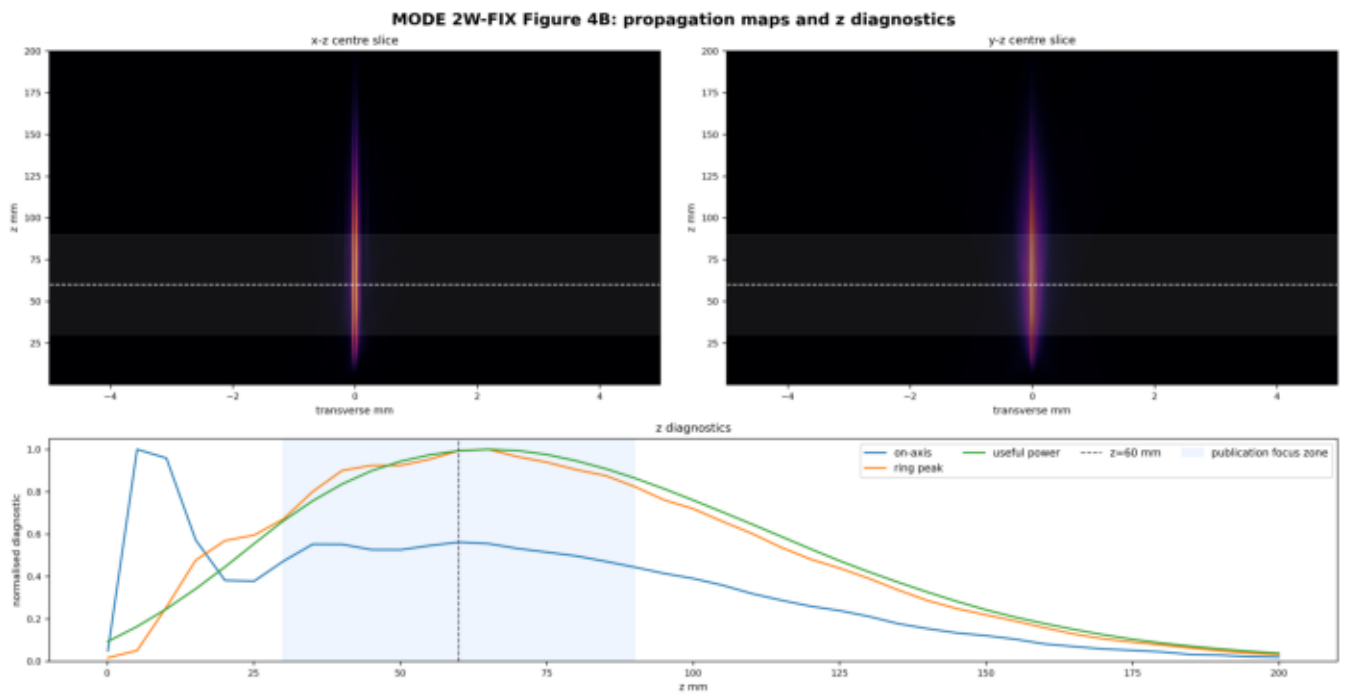
F11: Corrected recovery from a 0.5 mm axicon/mask offset.



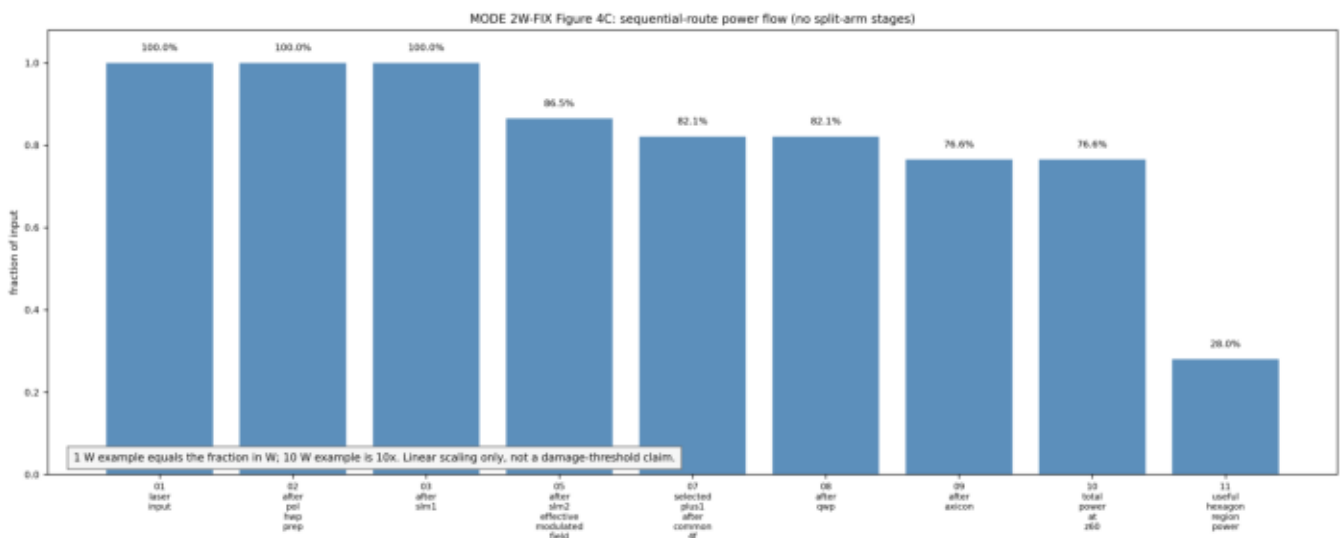
F12: Transverse propagation with physical context and SAS-scaled focus crops.



F13: x-z and y-z propagation maps plus z diagnostics.



F14: Sequential-route power flow.

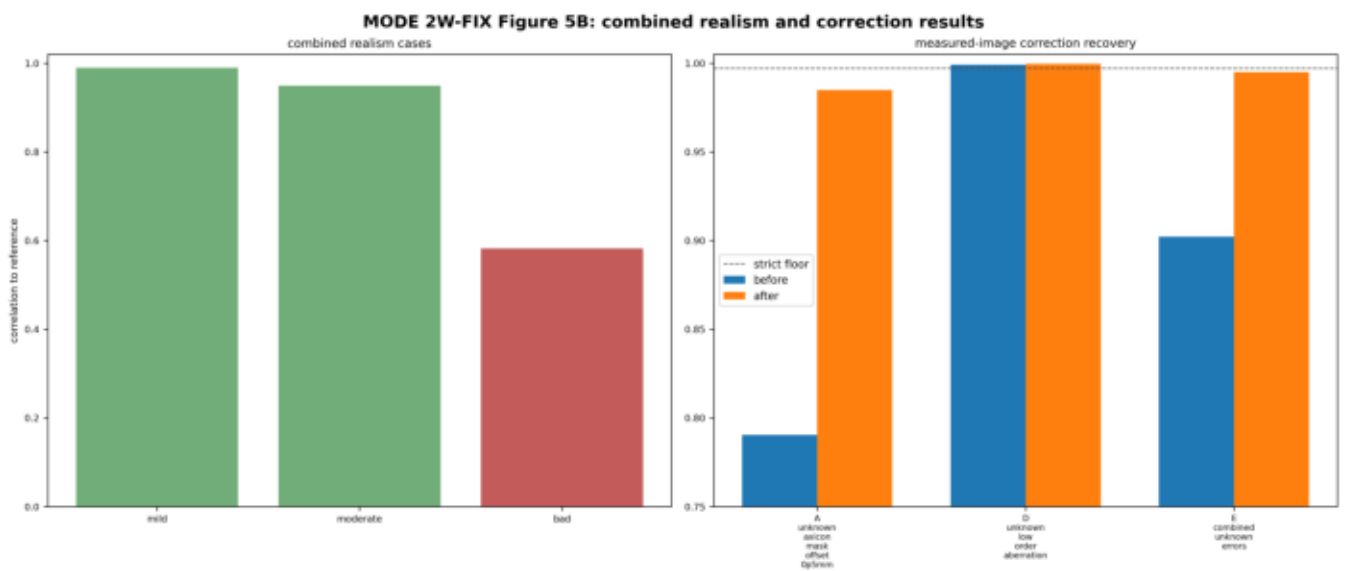


F15: Tolerance limits table/plot.

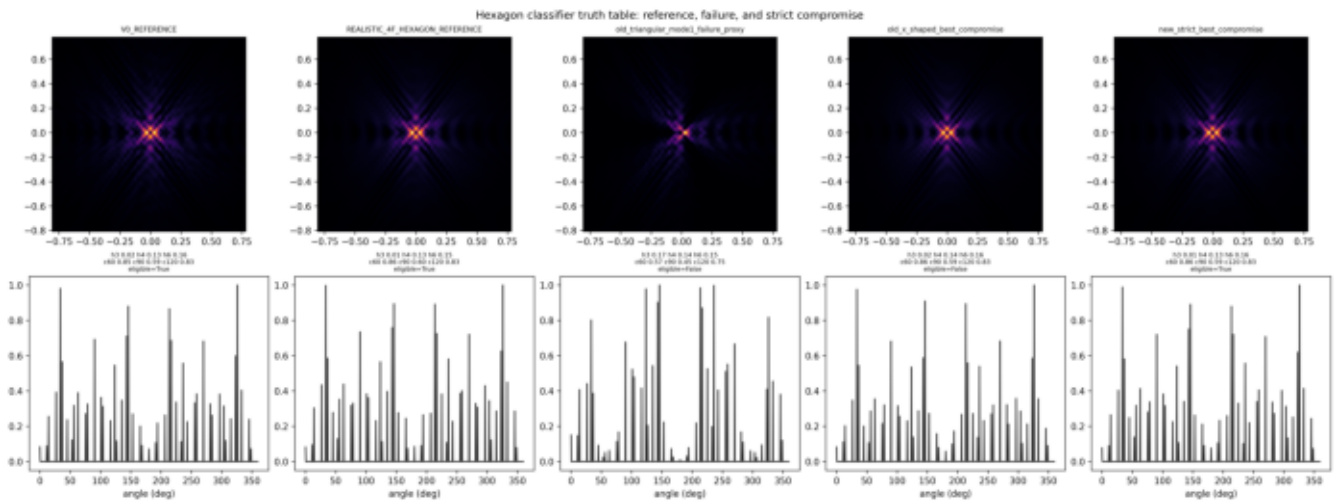
MODE 2W-FIX Figure 5A: tolerance limits, not pass-fraction bars

parameter	tested range	worst passing	first failure	min com	strict class
phase quantisation	16 to 1024	1024	not reached	0.9930	visual_hexagonal_field
M factor	0.93 fixed validated hardware-realism setting	0.93	not reached		visual_hexagonal_field at fixed setting
H/V imbalance	0.8 to 1.2	1.2	not reached	0.9827	visual_hexagonal_field
QWP angle	-2 to 2	-2	not reached	0.9931	visual_hexagonal_field
QWP retardance	-5 to 5	-5	not reached	0.9931	visual_hexagonal_field
iris radius	0.24 to 0.8	0.8	not reached	0.9844	visual_hexagonal_field
iris decentre	-1.5 to 1.5	-1.5	not reached	0.9869	visual_hexagonal_field
H/V registration	0 to 360	160	not reached	0.9753	visual_hexagonal_field
axicon decentre	0 to 1	0.2	0.5	0.7410	triangular_lobe_field, visual_hexagonal_field
z offset	-20 to 20	-20	not reached	0.9791	visual_hexagonal_field

F16: Combined realism and correction results.

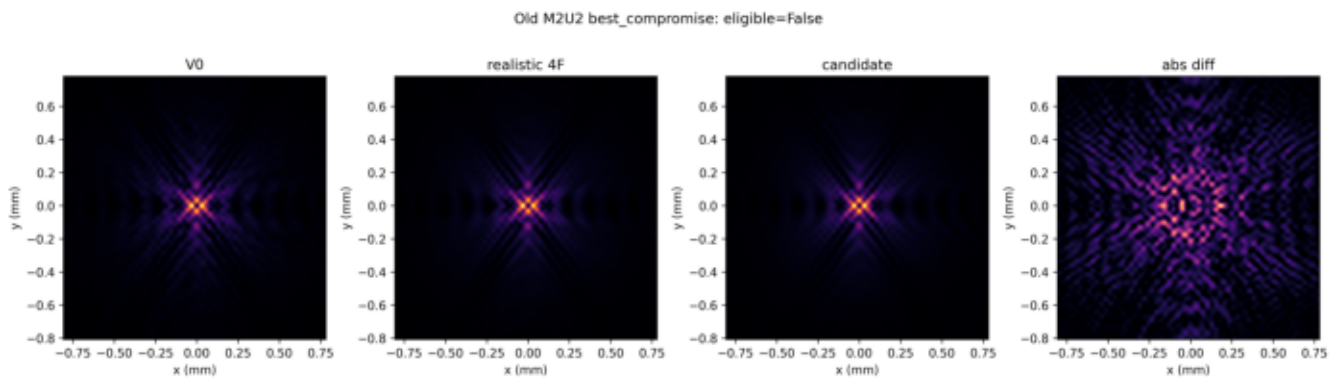


F17: Strict classifier truth table and gate calibration.



Source: outputs/figures/digital_twin/nathan_mode2u2_fix_strict_hexagon_optimisation/hexagon_classifier_truth_table_highres.png

S01: Forbidden old best-compromise audit figure.



Source:

outputs/figures/digital_twin/nathan_mode2u2_fix_strict_hexagon_optimisation/01_old_optima_audit/best_compromise_m2u2_opt_003_c5